Do the extra data improve our predictions?

Updated analysis: posterior simulation + source-by-source tests on the six existing terms.

What I tested

Grades alone, then LMS activity, EVC, wellness, every combination, and each source removed from the full model.

Main finding

No added average-error benefit beyond grades shown.

Simulation helps describe uncertainty. Later-term comparisons test predictive value.

191 students, six terms, four types of inputs

One row per student; Fall 2024–Spring 2026. Current analysis uses the saved processed table.

Grades

10-score or 16-score bundle
Release dates unverified

LMS activity

Log page views
Log participations

EVC

Expectancy, value
Source cost/support field

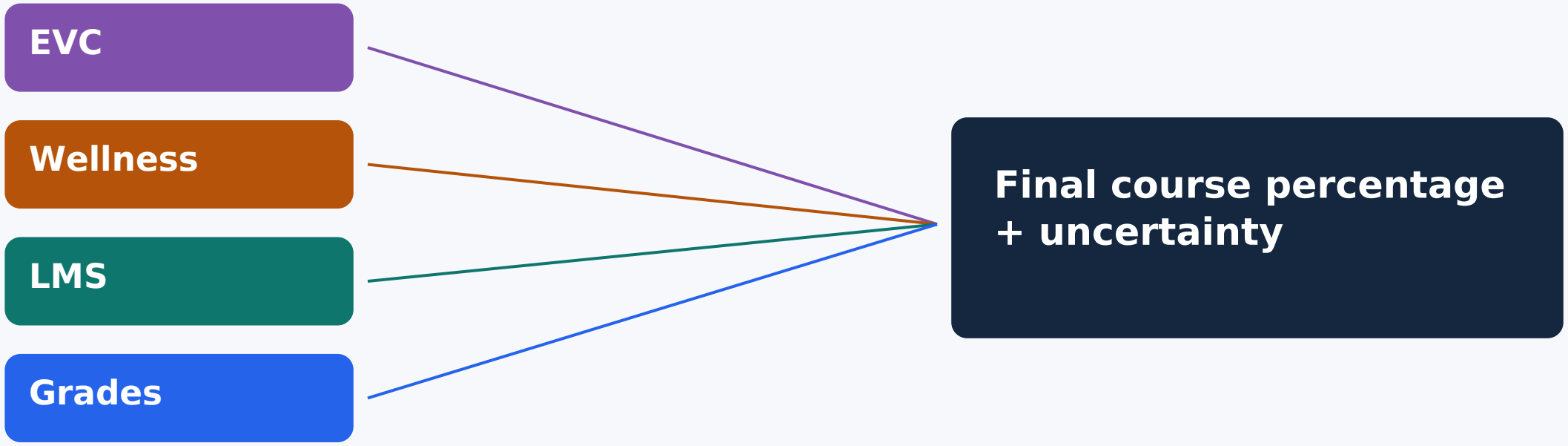
Wellness

Early self-report
Student mean

78/191 lack early surveys. The separate MLC RETRO export was unavailable and was not analyzed.

A useful map does not establish predictive value

The original fixed DAG is preserved. Its arrows encode modeling assumptions.



Grouped outcome-parent view only; original predictor-to-predictor arrows are omitted here.

Three different questions need three different checks

A likely positive coefficient is not the same as improved prediction.

1 Relationship

How plausible is a positive or practically large weight?

Conditional coefficient posterior.

2 Outcome

How likely is a student to finish below 70%?

Posterior predictive probability.

3 Added value

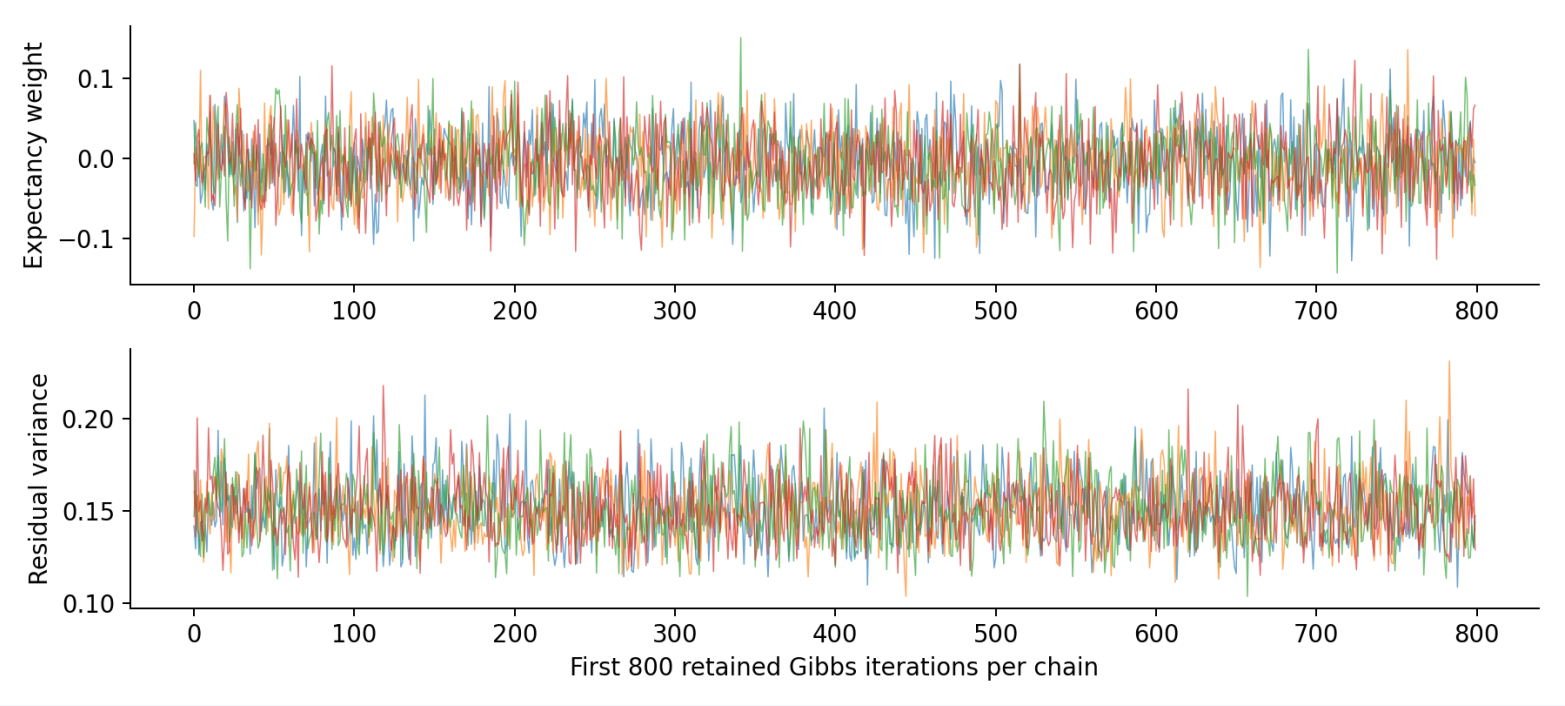
Do later students receive better predictions?

Paired errors + term bootstrap.

These are associations and forecasts; none establishes a coaching effect.

The sampler agrees with the exact Bayesian solution

The present linear Gaussian model has a closed-form posterior; MCMC is a validation exercise here.



4 chains

16,000 retained Gibbs draws

R-hat ≤ 1.0007

Minimum bulk ESS: 12,194
Exact means within 2.01 MCSE

Outcome conditional model only; this is not a sampler for the full missing-data DAG.

Train on earlier terms; predict the next term

Four checks, 146 later-student predictions per model. Every model uses the same evaluation students.

Learn from	Check on	Students
Fall 2024 + Winter 2025	Spring 2025	43
Through Spring 2025	Fall 2025	20
Through Fall 2025	Winter 2026	40
Through Winter 2026	Spring 2026	43

Three contexts: no grades / Block A (10 scores) / Block B (16 scores).

Retrospective evidence bundles—not verified week-4 forecasts. All findings remain exploratory.

Adding all three sources did not meaningfully reduce error

Macro MAE: average of the four term errors, in final-percentage points. Lower is better.

Available grades	Baseline	+ LMS + EVC + wellness	MAE reduction
No grades	6.54	6.52	+0.014
Block A	3.61	3.70	-0.089
Block B	2.95	3.04	-0.088

Block B: all-source MAE worsened in 4 of 4 later-term checks.

LMS shows a modest signal when grades are absent

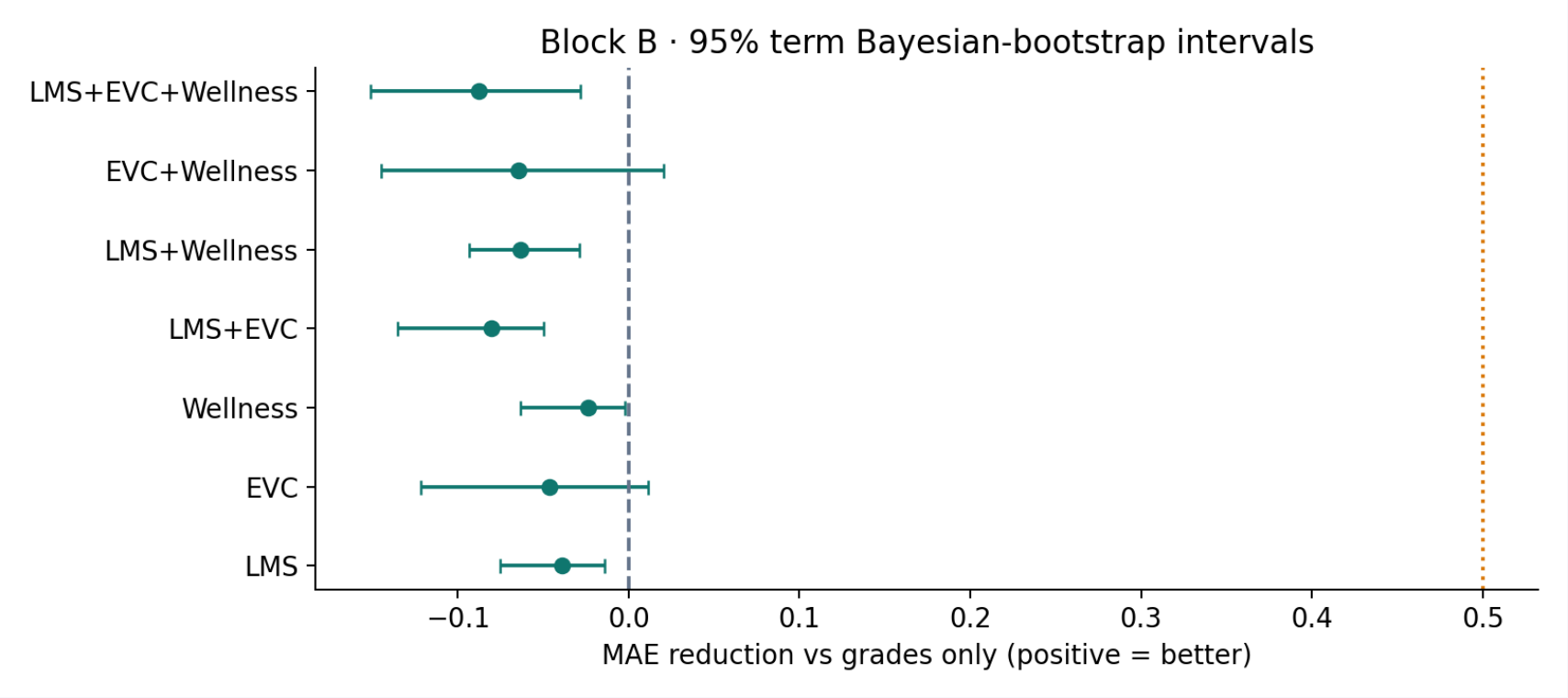
Exploratory source checks: no grade inputs; the same four later terms and students.

Inputs beyond mean baseline	Macro MAE	MAE reduction	Terms better
LMS activity	6.36	+0.177	4/4
Wellness	6.56	-0.020	2/4
EVC	6.71	-0.173	0/4
LMS + wellness	6.34	+0.198	4/4
All three sources	6.52	+0.014	2/4

Mean baseline MAE = 6.54. No-grades scenarios do not establish a verified prediction date.

No single addition shows a stable Block B MAE gain

Positive means a reduction in error relative to the same grades-only baseline.



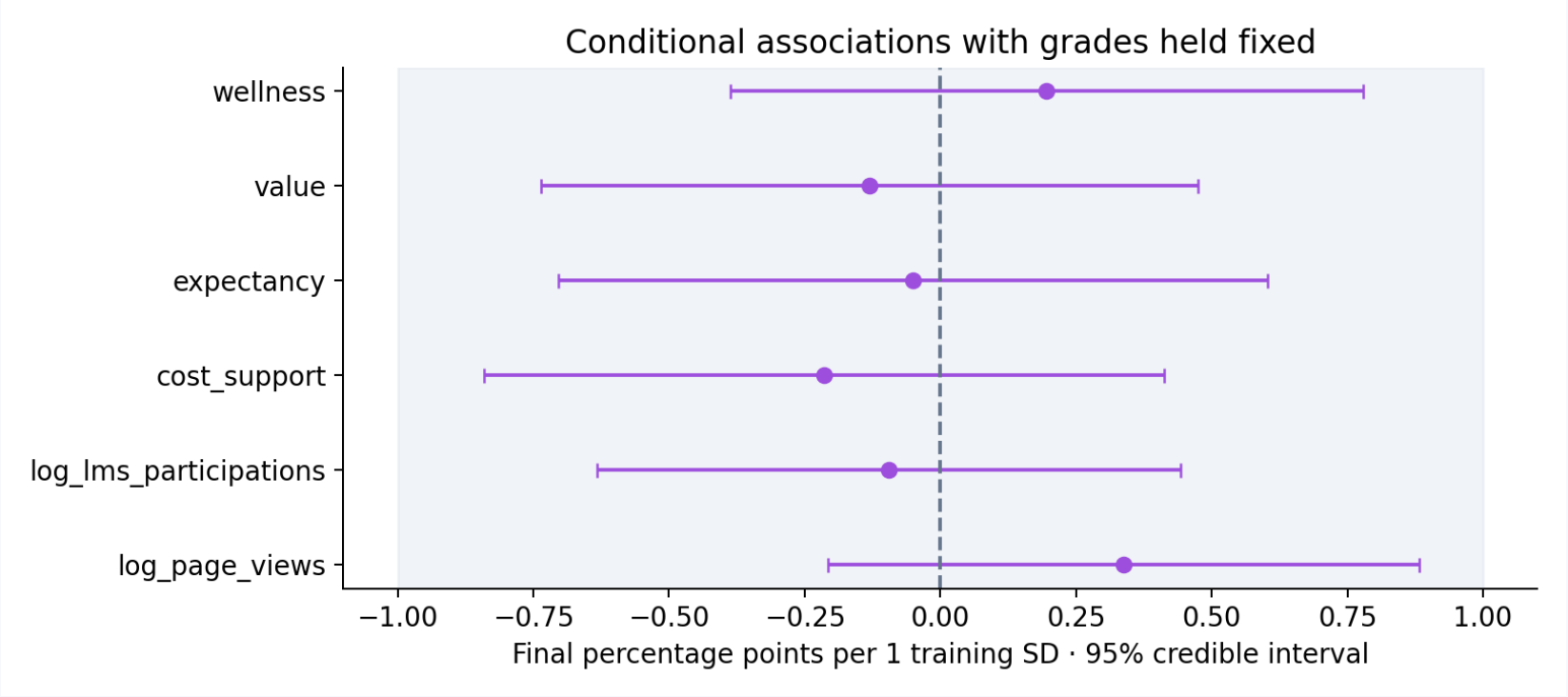
Interpretation

Changes are small. Do not treat four terms as strong evidence about every future class.

Intervals: 95% term Bayesian bootstrap; dotted line = illustrative 0.5-point useful gain.

Small conditional associations after grades are included

Final fit: earlier 148 students. Effects are percentage points per one training standard deviation.



Page views

P(weight > 0): 88.9%
Mean: +0.34 points / SD
P(|weight| > 1): 0.88%

All six non-grade P(|weight| > 1 point) values are below 1% in this conditional model.

A probability is useful—and can still be wrong

Illustrative question: P(final percentage < 70). This is not an official intervention threshold.

Example with complete inputs

Predicted mean: 71.24%
90% interval: 65.72–76.75%
P(final < 70): 35.6%
Recorded result: 75.95%

A low outcome the model missed

Predicted mean: 77.60%
90% interval: 71.36–83.83%
P(final < 70): 2.3%
Recorded result: 68.03%

The missed case had missing surveys. Imputed values are treated as fixed, so uncertainty is incomplete.

The diagnosis survives several reasonable alternatives

Gain from adding all three sources to Block B; positive is better.

Alternative	Block B MAE reduction
Main: precision 10	−0.088
Weaker shrinkage: precision 1	-0.191
Stronger shrinkage: precision 100	-0.007
Add missingness flags	-0.119
Complete non-grade respondents	-0.106

No-grades gains change with the prior; respondent-only results apply to a selected population.

Good exploratory framework; predictive claims need restraint

I would keep the Bayesian baseline and improve the data and likelihood before making the model more complex.

Keep

Transparent assumptions
Temporal evaluation
Exact Bayesian baseline
Posterior uncertainty

Improve

Grade availability
Survey scoring / response
Low-outcome fit
Missing-data uncertainty

Do not conclude

Arrows prove causality
MCMC guarantees accuracy
Extra surveys never help
Coaching benefit shown

Use the existing records to answer a sharper question

A new class can strengthen confirmation later; it is not required to finish this exploratory capstone.

1

Recover dates and measurement definitions

Grade release/revision snapshots; early EVC versus separate MLC RETRO.

2

Choose one useful forecast checkpoint

Use only inputs available then; retain the same students in comparisons.

3

Agree on what improvement would matter

The 0.5-point gain and 70% risk threshold used here are illustrative.

4

Address specific model failures

Test robust tails, term pooling or missing-data models against this baseline.

One posterior; two sampling routes

Normal–Inverse-Gamma regression, conditional on training-only preprocessing and imputed inputs.

$$z = A\beta + \varepsilon;\ \varepsilon \sim \text{Normal}(0, \sigma^2 I)$$
$$\beta \mid \sigma^2 \sim \text{Normal}(0, \sigma^2 \Lambda_0^{-1});\ \sigma^2 \sim \text{InvGamma}(2, 1)$$
$$L = \Lambda_0 + A^T A;\ m = L^{-1} A^T z;\ a = 2 + n/2$$
$$b = 1 + \tfrac{1}{2}(z^T z - m^T L m)$$
Direct: $\sigma^2 \sim \text{IG}(a,b)$, then $\beta \mid \sigma^2 \sim \text{Normal}(m, \sigma^2 L^{-1})$

Gibbs variance step: $\text{IG}(a+p/2,\ b+\tfrac{1}{2}(\beta-m)^T L(\beta-m))$

New outcome: Student-t with $df = 2a$, location $A^* m$, $\text{scale}^2 = (b/a)(1+A^* L^{-1} A^{*T})$

Parameters above are in standardized units; predictions are transformed back to final-percentage points.

Evaluate distributions as well as point errors

Four-term macro results for Block B. The 70% Brier threshold is illustrative.

Metric	Grades only	All sources
MAE ↓	2.9485	3.0364
Negative log density ↓	2.7606	2.7735
90% interval coverage	0.9061	0.9110
Mean interval width ↓	11.6881	12.0794
Brier score below 70 ↓	0.0591	0.0589

Near-nominal coverage is not proof of calibrated low-outcome probabilities.

What was run, and what remains unverified

All new tables, plots, code and executed notebook accompany this presentation.

Reproducible: MLC_Posterior_Predictive_Value.ipynb + posterior_study/
Evidence: posterior_results/ + POSTERIOR_STUDY.md
Preserved: original DAG notebook, original results, and supplied presentation.

Unverified: raw-data extraction; separate RETRO effects; grade timing; causal effects; prospective calibration.

Method references
Murphy (2012), Machine Learning: A Probabilistic Perspective, Bayesian linear regression.
Rubin (1981), The Bayesian Bootstrap. DOI: 10.1214/aos/1176345338.
Vehtari et al. (2021), Rank-Normalization, Folding, and Localization. DOI: 10.1214/20-BA1221.